

Exact Type Beyond Log-Convexity

A complete reversion criterion, sharp distortion,
and dimension-free nonlinear calculus

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Abstract

For an arbitrary positive coefficient weight $N = (N_n)$, we characterize exact preservation of exponential coefficient type under every formal compositional inverse. The criterion is superexponential root growth *together* with subexponential agreement with the least supermultiplicative majorant

$$S_n = \max_{j_1 + \dots + j_k = n} \prod_{i=1}^k N_{j_i}.$$

The central estimate is stronger: whenever $N_n^{1/n} \rightarrow \infty$, the positive extremal inverse has coefficient $S_n \exp(o(n))$, for every fixed positive input amplitude. Consequently $\limsup (S_n/N_n)^{1/n}$ is the exact optimal multiplicative distortion of type, not merely an upper bound.

For the zero-loss weights we prove a sharp composition inequality for two divergent arguments, determine all possible cancellations at equal type, and extend inversion and composition to formal Banach-space maps without a dimension-dependent loss. Explicit oscillating weights show that log-convex approximation is not necessary; other examples realize every finite distortion and even send type zero to infinite type under inversion. The proofs are self-contained after elementary formal-series algebra. An exact finite verification program, a source audit, and a further research program accompany the article.

Status and scope. Conventional mathematical proofs addressing explicit questions in ProveIt’s *Sharp Weighted Type Under Formal Reversion*. Not Lean-verified or independently peer-reviewed. The literature comparison is targeted; global publication priority is not asserted. These are formal coefficient theorems, not analytic summability theorems.

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Reading guide. The central argument is Sections 3–5. Section 6 gives the complete scalar type calculus; Section 7 specifies the norms needed for its vector-valued extension. The examples in Section 8 separate the exact criterion from several tempting regularity assumptions. Verification, formalization, and genuinely remaining questions are treated separately.

1 Research target and relation to prior work

1.1 Three questions from the repository

The ProveIt article *Sharp Weighted Type Under Formal Reversion* proves exact preservation of a weighted coefficient limsup for positive log-convex weights with superexponential roots. Its final research section asks for a necessary-and-sufficient criterion for arbitrary positive weights, a sharp rule when both composition arguments are divergent, and a multivariate or operator-valued extension [1]. Those are the specific targets of the present article. We do not revisit the repository’s quadratic-feedback coefficient asymptotics, inverse-digamma optimal truncation, or Hahn resonance calculations.

The arbitrary-weight question is more than a technical relaxation. Nonlinear inversion can multiply coefficients whose degrees add to the requested excess degree. A weight may be large at the input degrees and have a very deep dip at the output degree. A root-growth condition alone cannot exclude this failure. Conversely, oscillation need not be harmful: the products may still fit the target weight with no exponential loss. The correct condition concerns all finite additive partitions, not convexity of a sequence of logarithms.

We answer the first question completely for scalar formal power series. We also give a complete rule based only on the two scalar types: unequal types force equality in a maximum bound, while every smaller output type can occur at equal input type. For formal maps tangent to the identity on a Banach space, the same exact statements hold with multilinear operator norms. For a general invertible derivative we prove sharp two-sided norm bounds and explain why the derivative and one scalar input type cannot determine a single exact output type.

1.2 What is classical, and what is established here

The Lagrange inversion formula and its finite coefficient expansions are classical; Gessel’s survey is an appropriate reference [2]. We include the short formal argument needed here and make no novelty claim about formal invertibility, the coefficient formula, or scalar majorant methods.

Rainer and Schindl characterize composition in ultradifferentiable classes [3], and characterize equivalences among composition, inversion, and differential-equation stability under stated regularity hypotheses [4]. Their sequence formulation uses weak log-convexity, derivation conditions, and a Faà di Bruno bound allowing a factor C^n . Their discussion also credits earlier Banach-space inverse-function results. Thus neither nonlinear class stability nor Banach-space inversion is a new concept here.

The present question has a different precision and domain. We work with arbitrary *formal coefficient weights*, retain the exact exponential limsup constant, and allow neither an unspecified exponential loss nor an implicit convexity or derivation hypothesis. We prove that one explicit supermultiplicative envelope measures the entire worst-case loss. The upper estimate does not assume that the envelope is close to the original weight. This allows a classification of failure as well as success, and makes the subsequent composition and vector-valued results consequences of a single scalar estimate.

The proposed contributions are the envelope asymptotic, the complete arbitrary-weight criterion and optimal distortion theorem, the resulting exact scalar type calculus, and the explicit separating examples. Their proofs are supplied below. A targeted comparison is not an exhaustive priority search, and the assertions of this article should be independently checked before being treated as established literature.

1.3 Snapshot and dependency boundary

The repository checkpoint used for provenance is

3d5973524506411392a911470b5ddc35521568ea.

The principal source is

Analysis/Transseries/docs/series-and-transseries/
Sharp_Weighted_Type_Formal_Reversion/article.tex.

Its Git blob is

c2d9978806e0861f1bab7108b72ad92773e6a05c.

In particular, the pinned source's research questions on arbitrary weights, divergent composition, and multivariate inversion were read [1]. Selected neighboring package descriptions were inspected to avoid obvious overlap. The whole repository, the canonical large transseries volume, and newly arrived binary research archives were not audited claim by claim. No upstream file was modified.

No mathematical theorem from an unverified repository article is needed as a premise of the proofs below. The log-convex result is recovered independently in Proposition 5.5. No proof assistant was run; the attached computations are finite checks, not substitutes for the quantified arguments.

2 Weights, excess degree, and the main statements

2.1 The exact type being measured

Throughout, a weight means a sequence

$$N_0 = 1, \quad N_n \in (0, \infty) \quad (n \geq 1).$$

It need not be increasing, log-convex, weakly log-convex, or stable under a shift of its index. Let

$$f(z) = z + \sum_{n \geq 1} f_{n+1} z^{n+1} \in \mathbb{C}[[z]] \quad (2.1)$$

be tangent to the identity. Its *excess-degree type* is

$$T_N(f) = \limsup_{n \rightarrow \infty} \left(\frac{|f_{n+1}|}{N_n} \right)^{1/n} \in [0, \infty]. \quad (2.2)$$

The excess degree of z^{n+1} is n . This indexing is natural because excess degrees add in the inverse coefficient formula. We write $f^{(-1)}$ for the compositional inverse and reserve $1/f$ for a multiplicative reciprocal.

To translate the usual coefficient weight M_n , put $N_n = M_{n+1}$ for $n \geq 1$. Replacing the exponent $1/n$ in (2.2) by $1/(n+1)$ does not change the extended-real limsup. Thus the convention measures exactly the same type as the repository's ordinary-degree convention, with the harmless finite linear coefficient omitted.

For $A > 0$, define the tangent dilation

$$(\mathcal{D}_A f)(z) = A f(z/A). \quad (2.3)$$

Then

$$T_N(\mathcal{D}_A f) = A^{-1} T_N(f), \quad (\mathcal{D}_A f)^{(-1)} = \mathcal{D}_A(f^{(-1)}). \quad (2.4)$$

If $T_N(f) < A$, there is a finite $C > 0$ such that

$$|f_{n+1}| \leq CA^n N_n \quad (n \geq 1). \quad (2.5)$$

This follows directly from the definition of limsup, increasing C to cover the finitely many exceptional coefficients.

Definition 2.1 (Envelope and distortion). The *supermultiplicative envelope* of N is $S_0 = 1$ and

$$S_n = \max_{\substack{k \geq 1, j_1, \dots, j_k \geq 1 \\ j_1 + \dots + j_k = n}} \prod_{i=1}^k N_{j_i} \quad (n \geq 1). \quad (2.6)$$

Define

$$\Delta(N) = \limsup_{n \rightarrow \infty} (S_n/N_n)^{1/n} \in [1, \infty]. \quad (2.7)$$

We call N *admissible* if

$$N_n^{1/n} \rightarrow \infty, \quad \log(S_n/N_n) = o(n). \quad (2.8)$$

There are only finitely many ordered compositions of each positive integer, so (2.6) is a finite maximum even for arbitrary real weights. Positivity and finite support at each coefficient are all that is required for the formal manipulations.

2.2 The classification and its quantitative refinement

Theorem 2.2 (Complete zero-loss classification). *For every positive weight N , the following are equivalent.*

- (i) *Every tangent-to-identity scalar formal series satisfies $T_N(f^{(-1)}) = T_N(f)$, including types zero and infinity.*
- (ii) *N is admissible in the sense of (2.8).*
- (iii) *There exists a positive supermultiplicative sequence V , with $V_0 = 1$, $V_n^{1/n} \rightarrow \infty$, and $\log(V_n/N_n) = o(n)$.*

The equivalence is proved in Section 5. The next statement quantifies how it fails. Let $\mathcal{I}(N)$ be the infimum of all finite $D \geq 1$ such that

$$T_N(f^{(-1)}) \leq DT_N(f) \quad \text{for every tangent-to-identity } f. \quad (2.9)$$

The infimum of the empty set is infinity; for finite D we use $D \cdot 0 = 0$ and $D \cdot \infty = \infty$.

Theorem 2.3 (Exact optimal distortion). *For an arbitrary positive weight,*

$$\mathcal{I}(N) = \begin{cases} \Delta(N), & N_n^{1/n} \rightarrow \infty, \\ \infty, & N_n^{1/n} \not\rightarrow \infty. \end{cases} \quad (2.10)$$

When $\Delta(N) < \infty$ and the roots tend to infinity,

$$\frac{T_N(f)}{\Delta(N)} \leq T_N(f^{(-1)}) \leq \Delta(N)T_N(f). \quad (2.11)$$

The upper constant is attained by the explicit series $f = z - \sum_{n \geq 1} N_n z^{n+1}$, whose input type is one.

Thus there is a canonical obstruction, not just a list of sufficient regularity conditions. For instance, a weight may have roots tending to infinity yet have $\Delta(N) = 2$, or $\Delta(N) = \infty$. Both possibilities will be realized by elementary explicit formulas.

2.3 The positive extremal series

For a fixed $C > 0$, put

$$W_N(z) = \sum_{n \geq 1} N_n z^{n+1}, \quad H_C(z) = (z - CW_N(z))^{\langle -1 \rangle} = z + \sum_{n \geq 1} h_{n+1}(C) z^{n+1}. \quad (2.12)$$

It is equivalently the unique formal solution of

$$H_C = z + CW_N(H_C). \quad (2.13)$$

All its coefficients are nonnegative. The following is the technical core of the article.

Theorem 2.4 (Envelope asymptotic). *If $N_n^{1/n} \rightarrow \infty$, then for every fixed $C > 0$,*

$$\boxed{\lim_{n \rightarrow \infty} \left(\frac{h_{n+1}(C)}{S_n} \right)^{1/n} = 1.} \quad (2.14)$$

No bound on S_n/N_n is assumed.

The assertion is about an exponential scale, not an asymptotic equivalent $h_{n+1}(C) \sim S_n$. The residual ratio can grow or decay subexponentially and may oscillate. Both the uniformity in the coefficient index and the fact that C may be smaller than one matter in the proof.

3 Envelope algebra and finite inverse coefficients

3.1 A least supermultiplicative majorant

Lemma 3.1 (Envelope calculus). *For every positive N , its envelope satisfies*

$$S_n \geq N_n, \quad S_{a+b} \geq S_a S_b \quad (a, b \geq 0), \quad S_n = \max_{1 \leq j \leq n} N_j S_{n-j}. \quad (3.1)$$

It is the least pointwise supermultiplicative majorant of N with zeroth term one. It is unchanged if ordered compositions in (2.6) are replaced by unordered integer partitions.

Proof. The one-part composition proves $S_n \geq N_n$. Concatenating maximizing compositions of a and b proves supermultiplicativity. Removing the first part of a composition, and conversely adjoining a first part, gives the recurrence. If $V \geq N$ and $V_{a+b} \geq V_a V_b$, repeated multiplication gives $V_n \geq \prod_i V_{j_i} \geq \prod_i N_{j_i}$ for every composition of n ; hence $V_n \geq S_n$. Finally, the product depends on the multiplicities of parts, not their order. $\square$

Writing $s_n = \log S_n$ and $a_n = \log N_n$ converts the recurrence to

$$s_0 = 0, \quad s_n = \max_{1 \leq j \leq n} (a_j + s_{n-j}). \quad (3.2)$$

This is a finite max-plus closure. The obstruction to zero loss is the excess $s_n - a_n$ on a linear scale. It is not the second difference of a_n , nor a bound on a single adjacent quotient.

3.2 Formal inversion and a residue calculation

Every series of the form (2.1) has a unique formal inverse: the coefficient of degree m in $f(g(z)) = z$ determines g_m using only earlier coefficients. Associativity then gives a two-sided inverse. All operations up to a prescribed degree use finitely many terms.

For completeness, the needed Lagrange identity can be obtained directly from formal residues. If $g = f^{(-1)}$ and $m \geq 1$, substitution $z = f(t)$ in a residue gives

$$\begin{aligned} [z^m]g(z) &= \text{Res}_z g(z) z^{-m-1} dz \\ &= \text{Res}_t t f(t)^{-m-1} f'(t) dt = \frac{1}{m} \text{Res}_t f(t)^{-m} dt. \end{aligned}$$

The last equality follows because the residue of a formal derivative vanishes, applied to $d(t f(t)^{-m})/dt$. Formal substitution is legitimate since $f(t) = t$ times a unit. Therefore

$$[z^m]g(z) = \frac{1}{m} [t^{m-1}] \left(\frac{t}{f(t)} \right)^m. \quad (3.3)$$

This is the classical formula used below [2].

Lemma 3.2 (Positive coefficient formula and absolute majorant). *For (2.12) and $n \geq 1$,*

$$h_{n+1}(C) = \frac{1}{n+1} \sum_{k=1}^n \binom{n+k}{k} C^k \sum_{\substack{j_1+\dots+j_k=n \\ j_i \geq 1}} \prod_{i=1}^k N_{j_i}. \quad (3.4)$$

Equivalently, summing over finitely supported multiplicity vectors $\mathbf{m} = (m_1, m_2, \dots)$ with $\sum j m_j = n$ and $k = \sum m_j$,

$$h_{n+1}(C) = \sum_{\mathbf{m}} \frac{(n+k)!}{(n+1)! \prod_j m_j!} C^k \prod_j N_j^{m_j}. \quad (3.5)$$

If $|f_{n+1}| \leq C N_n$ for all n , then

$$|[z^{n+1}]f^{(-1)}| \leq h_{n+1}(C). \quad (3.6)$$

Proof. In (3.3), take $m = n+1$ and expand

$$\left(1 - C \sum_{j \geq 1} N_j t^j \right)^{-n-1} = \sum_{k \geq 0} \binom{n+k}{k} C^k \left(\sum_{j \geq 1} N_j t^j \right)^k.$$

Only $1 \leq k \leq n$ contributes to degree n . Grouping the $k!/\prod_j m_j!$ orderings of a fixed partition gives (3.5). For a signed or complex input, the same finite expansion and the triangle inequality give (3.6). $\square$

Remark 3.3 (A coefficient lower bound). For any ordered composition of n with k parts,

$$h_{n+1}(C) \geq C^k \prod_i N_{j_i}, \quad (3.7)$$

because $\binom{n+k}{k}/(n+1) \geq 1$. In particular, $h_{n+1}(1) \geq S_n$. A maximizing composition need not be unique; the next lemma controls the lengths of all maximizers.

4 Sparse products, dense suppression, and the central limit

4.1 Maximizing partitions have sublinear length

Lemma 4.1 (Sparse maximizers). *Assume $N_n^{1/n} \rightarrow \infty$. For every $\eta > 0$, all compositions attaining S_n have fewer than ηn parts once n is sufficiently large. Equivalently, the maximum number of parts in a maximizing composition is $o(n)$.*

Proof. It suffices to consider $0 < \eta \leq 1$. Set $L = \lceil 2/\eta \rceil$ and

$$B = \max \left(1, \max_{1 \leq j \leq L} N_j^{1/j} \right).$$

Suppose a composition of n has $k \geq \eta n$ parts. At most $n/(L+1) < \eta n/2$ parts exceed L . Thus more than $\eta n/2$ parts are at most L . Let ℓ be their total mass. Then $\ell \geq \eta n/2$, and their product is at most B^ℓ .

Since $N_\ell^{1/\ell} \rightarrow \infty$, for all sufficiently large n we have $N_\ell > B^\ell$ whenever $\ell \geq \eta n/2$. Replace those small parts by the single part ℓ . This preserves the sum n and strictly increases the product. Consequently the original composition was not maximizing. The threshold is independent of the chosen composition. $\square$

This argument does not assume monotone roots. The full root limit makes the replacement by one part valid uniformly beyond a large index. A deep dip could prevent that particular one-part replacement. A strengthened version, using a whole maximizing subpartition instead, is given at the end of this section.

4.2 A finite upper certificate

Define for $1 \leq k \leq n$

$$A_{n,k} = \frac{1}{n+1} \binom{n+k}{k} \binom{n-1}{k-1}. \quad (4.1)$$

The second binomial counts ordered compositions of n with k positive parts. For $0 < \eta \leq 1/2$ and $C > 0$, set

$$P_{n,\eta}(C) = \sum_{1 \leq k \leq \eta n} A_{n,k} C^k, \quad Q_{n,\eta}(C) = \sum_{\eta n < k \leq n} A_{n,k} C^k. \quad (4.2)$$

A sum with no indices is zero.

Theorem 4.2 (Exact finite sparse–dense bound). *Fix any positive weight, $n \geq 1$, $C > 0$, and $0 < \eta \leq 1/2$. Let $L = \lceil 2/\eta \rceil$ and choose $B \geq 1$ satisfying $N_j \leq B^j$ for $1 \leq j \leq \min(L, n)$. Put*

$$R_{n,\eta,B} = \max_{\lceil \eta n/2 \rceil \leq \ell \leq n} \frac{B^\ell}{S_\ell}. \quad (4.3)$$

Then

$$\boxed{h_{n+1}(C) \leq S_n(P_{n,\eta}(C) + Q_{n,\eta}(C)R_{n,\eta,B})}. \quad (4.4)$$

No limiting hypothesis on N is needed for this inequality.

Proof. For compositions with $k \leq \eta n$, each product is at most S_n ; these give the first term. If $k > \eta n$, separate the parts at most L . Their number exceeds $\eta n/2$, as in Lemma 4.1, and their total mass ℓ satisfies $\ell \geq \lceil \eta n/2 \rceil$. Their product is at most B^ℓ . The product of the remaining parts is at most $S_{n-\ell}$, including the empty product $S_0 = 1$. Since $S_\ell S_{n-\ell} \leq S_n$, the whole product divided by S_n is at most $B^\ell/S_\ell \leq R_{n,\eta,B}$. Count compositions and apply (3.4). $\square$

All entries of (4.4) involve only the finite prefix $N_1, \dots, N_n$. For rational weights an integer B can be chosen by exact root comparisons, so this is an executable rational inequality rather than a certificate requiring floating-point logarithms.

4.3 Subexponential sparse entropy

Let $\mathfrak{h}(t) = -t \log t - (1-t) \log(1-t)$ for $0 < t < 1$, extended by zero at the endpoints. The elementary binomial entropy inequality is

$$\binom{r}{s} \leq \exp(r\mathfrak{h}(s/r)). \quad (4.5)$$

For $0 < s < r$, it follows by taking the s th term in $1 = (s/r + (1-s/r))^r$ and rearranging; the endpoint cases are immediate.

Lemma 4.3 (Counting bounds). *For $0 < \eta \leq 1/2$,*

$$P_{n,\eta}(C) \leq n \exp(nE_C(\eta)), \quad E_C(\eta) = 2\mathfrak{h}(\eta/2) + \mathfrak{h}(\eta) + \eta \log^+ C, \quad (4.6)$$

where $E_C(\eta) \rightarrow 0$ as $\eta \downarrow 0$. Also

$$Q_{n,\eta}(C) \leq n(8 \max(1, C))^n. \quad (4.7)$$

Proof. For $k \leq \eta n$, use $\binom{n+k}{k} \leq \binom{2n}{k}$ and $\binom{n-1}{k-1} = (k/n) \binom{n}{k} \leq \binom{n}{k}$. The function $\mathfrak{h}$ is increasing on $[0, 1/2]$, so (4.5) bounds the binomials by $\exp(2n\mathfrak{h}(\eta/2))$ and $\exp(n\mathfrak{h}(\eta))$, respectively. Also $C^k \leq \exp(n\eta \log^+ C)$, and there are at most n summands. For the second assertion use the cruder bounds $\binom{n+k}{k} \leq 2^{2n}$, $\binom{n-1}{k-1} \leq 2^n$, and $C^k \leq \max(1, C)^n$. $\square$

The sparse count has arbitrarily small exponential rate. The dense count can have a fixed exponential rate, but its products incur a stronger penalty from the weight. This separation is the reason no smoothness of the weight profile is required.

Lemma 4.4 (Dense suppression). *If $N_n^{1/n} \rightarrow \infty$, then for fixed $\eta > 0$ and fixed $B \geq 1$,*

$$R_{n,\eta,B}^{1/n} \rightarrow 0. \quad (4.8)$$

Proof. We have $S_\ell \geq N_\ell$. Given $K > B$, eventually $S_\ell^{1/\ell} \geq K$ for every $\ell \geq \lceil \eta n/2 \rceil$. Therefore

$$R_{n,\eta,B} \leq (B/K)^{\eta n/2}, \quad \limsup_n R_{n,\eta,B}^{1/n} \leq (B/K)^{\eta/2}.$$

Let $K \rightarrow \infty$. $\square$

4.4 Proof of the envelope asymptotic

Proof of Theorem 2.4. For each n , select a maximizing composition with k_n parts. By Lemma 4.1, $k_n/n \rightarrow 0$. The lower bound (3.7) gives

$$(h_{n+1}(C)/S_n)^{1/n} \geq C^{k_n/n} \rightarrow 1.$$

This works for every fixed $C > 0$, including $C < 1$.

For the upper bound, fix $0 < \eta \leq 1/2$, let $L = \lceil 2/\eta \rceil$, and choose once and for all $B \geq \max(1, \max_{1 \leq j \leq L} N_j^{1/j})$. Combining Theorem 4.2 and Lemmas 4.3 and 4.4 gives

$$\limsup_n (h_{n+1}(C)/S_n)^{1/n} \leq \max(e^{E_C(\eta)}, 0) = e^{E_C(\eta)}.$$

Here the n th root of the dense term tends to zero despite its fixed exponential counting factor. Letting $\eta \downarrow 0$ gives an upper bound of one, and proves (2.14). $\square$

Corollary 4.5 (Type of the extremal inverse). *If $N_n^{1/n} \rightarrow \infty$, then for every fixed $C > 0$,*

$$T_N(H_C) = \Delta(N), \quad T_N(z - CW_N) = 1. \quad (4.9)$$

Proof. Multiply $(h_{n+1}(C)/S_n)^{1/n} \rightarrow 1$ by $(S_n/N_n)^{1/n}$ and take limsups, allowing infinity. For the input, $C^{1/n} \rightarrow 1$. $\square$

4.5 A strengthening for unbounded root profiles

Proposition 4.6 (The envelope needs only unbounded primitive roots). *For a positive weight N , the following are equivalent:*

$$\sup_{n \geq 1} N_n^{1/n} = \infty, \quad S_n^{1/n} \longrightarrow \infty.$$

Under either condition the conclusion (2.14) holds for every fixed $C > 0$, even if $N_n^{1/n}$ itself does not tend to infinity.

Proof. If the primitive roots are bounded by K , every composition product is at most K^n , and $S_n \leq K^n$. Conversely, fix $m \geq 1$ and write $n = qm + r$ with $0 \leq r < m$. Supermultiplicativity gives $S_n \geq N_m^q S_r$, so $\liminf_n S_n^{1/n} \geq N_m^{1/m}$. Taking arbitrarily large primitive roots proves the equivalence.

In the proof of Lemma 4.1, the small parts have mass $\ell \geq \eta n/2$ and product at most B^ℓ . Now $S_\ell > B^\ell$ for all such ℓ once n is large. Replace those small parts by any composition attaining S_ℓ , rather than by the single part ℓ . This again strictly increases the product, so all maximizing compositions still have sublinear length. The lower bound for $h_{n+1}(C)/S_n$ follows unchanged. The finite upper certificate already holds for arbitrary weights, and the dense-suppression proof only needs $S_\ell^{1/\ell} \rightarrow \infty$. The entropy argument therefore proves (2.14) under the weaker hypothesis. $\square$

The full limit $N_n^{1/n} \rightarrow \infty$ remains necessary for universal zero loss, by the polynomial obstruction below. The strengthening concerns the extremal envelope asymptotic, not an exception to that classification. If the primitive roots are unbounded but have a bounded subsequence, $S_n^{1/n} \rightarrow \infty$ forces $\Delta(N) = \infty$.

5 Complete classification, optimal loss, and canonical repair

5.1 Why superexponential roots are necessary

Lemma 5.1 (The polynomial obstruction). *If a finite D satisfies (2.9), then $N_n^{1/n} \rightarrow \infty$.*

Proof. The polynomial $p(z) = z - z^2$ has type zero for every weight. Its inverse is

$$p^{\langle -1 \rangle}(z) = \frac{1 - \sqrt{1 - 4z}}{2} = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} z^{n+1}.$$

The central binomial coefficient satisfies $4^n/(2n+1) \leq \binom{2n}{n} \leq 4^n$: it is the largest of the $2n+1$ binomial coefficients whose sum is 4^n . Consequently the n th root of the coefficient of z^{n+1} tends to four. If $N_n^{1/n}$ does not tend to infinity, some subsequence has $N_n^{1/n} \leq B < \infty$, and $T_N(p^{\langle -1 \rangle}) \geq 4/B > 0$. This contradicts $T_N(p^{\langle -1 \rangle}) \leq DT_N(p) = 0$. $\square$

The obstruction concerns type zero, not just the growth of a generic divergent series. It explains why a criterion based solely on the envelope defect would be incomplete. For example, $N_n = 1$ is already supermultiplicative and has $S_n = N_n$, but it cannot preserve the type of an inverse polynomial.

5.2 The exact distortion constant

Proof of Theorem 2.3. If the roots do not tend to infinity, Lemma 5.1 rules out every finite D , as claimed. Assume henceforth that $N_n^{1/n} \rightarrow \infty$.

First suppose $\Delta = \Delta(N) < \infty$ and $T_N(f) < \infty$. For each $A > T_N(f)$, choose C in (2.5). The coefficients of $\mathcal{D}_A f$ are bounded in absolute value by $C N_n$. Thus Lemma 3.2 and Corollary 4.5 give

$$A^{-1} T_N(f^{(-1)}) = T_N((\mathcal{D}_A f)^{(-1)}) \leq T_N(H_C) = \Delta.$$

Let $A \downarrow T_N(f)$. In particular a type-zero series has a type-zero inverse. The upper bound is automatic when $T_N(f) = \infty$. It therefore holds for every f . Applying it to $f^{(-1)}$ proves the lower bound in (2.11); when $T_N(f) = \infty$, a finite inverse type would contradict that same inequality.

The input $z - W_N$ has type one and its inverse has type Δ by Corollary 4.5. Hence no smaller constant works. If $\Delta = \infty$, this same input rules out every finite constant. This proves the formula in all cases. $\square$

The result controls a limsup type, not the ratio of individual signed inverse coefficients to their inputs. Individual coefficients may cancel completely. The positive extremal series removes cancellation and identifies the optimal universal obstruction.

5.3 Zero loss and equivalence of weights

Two positive weights will be called *subexponentially equivalent* when

$$\log(V_n/N_n) = o(n). \quad (5.1)$$

They define the same exact type for every coefficient sequence: multiplying each normalized n th root by $(N_n/V_n)^{1/n} \rightarrow 1$ preserves every extended-real limsup.

Proof of Theorem 2.2. Assume (i). The polynomial obstruction gives superexponential roots. The input $z - W_N$ and Corollary 4.5 then give $\Delta(N) = 1$. Since $S_n/N_n \geq 1$, this is equivalent to $\log(S_n/N_n) = o(n)$, proving (ii).

If (ii) holds, Theorem 2.3 with $\Delta = 1$ proves (i), and $V = S$ proves (iii).

Finally suppose (iii). Since V is supermultiplicative, its own envelope is exactly V . Its roots tend to infinity, so it is admissible and the already proved implication (ii) $\Rightarrow$ (i) applies to V . Subexponential equivalence identifies the V -type and N -type of every series and its inverse. Hence (i) also holds for N . $\square$

Corollary 5.2 (Stability under subexponential perturbations). *If N is admissible and $\log(V_n/N_n) = o(n)$, then V is admissible. In particular, altering finitely many weights by arbitrary positive factors does not affect admissibility or any exact type.*

Proof. All exact types coincide for the two weights. Transfer the universal inversion identity and apply Theorem 2.2. $\square$

This proof deliberately uses the characterization rather than multiplying pointwise $\exp(o(n))$ errors over an arbitrary partition. A partition could contain linearly many small parts; such an unqualified multiplication would lose the very precision at issue.

5.4 A minimal admissible repair

Theorem 5.3 (Canonical envelope repair). *Suppose $N_n^{1/n} \rightarrow \infty$. Its envelope S is admissible. For any admissible pointwise majorant $V \geq N$,*

$$\limsup_{n \rightarrow \infty} (S_n/V_n)^{1/n} \leq 1. \quad (5.2)$$

Thus S is the least supermultiplicative majorant pointwise, and is a minimal admissible repair up to subexponential domination.

Proof. The sequence S is supermultiplicative and has roots tending to infinity, so its own envelope is S and it is admissible. Let S^V denote the envelope of V . Monotonicity of the maximum-product construction gives $S_n \leq S_n^V$. Since V is admissible, $(S_n^V/V_n)^{1/n} \rightarrow 1$, proving (5.2). $\square$

Remark 5.4 (The precise minimality claim). We do not claim that S is the pointwise smallest among all admissible majorants. An admissible weight may have a finite or subexponential defect from exact supermultiplicativity. The pointwise minimality applies to supermultiplicative majorants; (5.2) is the separate asymptotic statement for admissible ones.

5.5 Recovery of the log-convex theorem

Proposition 5.5 (Log-convex weights need no eventual repair). *Let $M_0 = 1$ and let M be positive and log-convex, with $M_m^{1/m} \rightarrow \infty$. Set $N_0 = 1$ and $N_n = M_{n+1}$ for $n \geq 1$. Then its envelope satisfies $S_n = N_n$ for every sufficiently large n . In particular N is admissible.*

Proof. Log-convexity says the quotients M_j/M_{j-1} are nondecreasing. For integers $a \geq b \geq 2$, transferring one unit from b to a therefore gives $M_a M_b \leq M_{a+1} M_{b-1}$ whenever $b \geq 3$. Repeatedly transfer until all but one indices equal two. For $j_1 + \cdots + j_k = n$, with $j_i \geq 1$, this gives

$$\prod_{i=1}^k M_{j_i+1} \leq M_{n-k+2} M_2^{k-1}. \quad (5.3)$$

Since $M_0 = 1$, convexity of $m \mapsto \log M_m$ implies $M_r \leq M_{n+1}^{r/(n+1)}$ for $0 \leq r \leq n+1$. Thus the right side of (5.3) is at most

$$M_{n+1} \left(\frac{M_2}{M_{n+1}^{1/(n+1)}} \right)^{k-1}.$$

For all sufficiently large n the parenthesis is at most one, uniformly in k . Every partition product is then at most M_{n+1} , and the one-part partition attains that value. Also $N_n^{1/n} = (M_{n+1}^{1/(n+1)})^{(n+1)/n} \rightarrow \infty$. $\square$

This is the log-convex special case treated in the predecessor [1], recovered here rather than imported as a new result. It also explains why the one-step shift in the definition of N is important: the products in inversion involve M_{j_i+1} , not M_{j_i} . The shifted weight itself need not satisfy a normalization-dependent log-convex condition at its first few indices.

6 Two divergent composition arguments and exact cancellation

In this section N is admissible. No convergence of a formal series in a nonzero disk is assumed.

6.1 The maximum bound

For nonnegative-coefficient series, write $F \preceq G$ when every coefficient of F is at most the corresponding coefficient of G . Substitution by series with zero constant coefficient preserves this order. Put $B_C = z + CW_N$.

Lemma 6.1 (A single inverse majorizes a double composition). *For every $C > 0$,*

$$B_C \circ B_C \preceq H_{2C}. \quad (6.1)$$

Proof. The identity $H_{2C} = z + 2CW_N(H_{2C})$ gives $z \preceq B_C \preceq H_{2C}$. Consequently

$$B_C(B_C) = z + CW_N(z) + CW_N(B_C) \preceq z + 2CW_N(H_{2C}) = H_{2C}.$$

□

Theorem 6.2 (Sharp composition law). *For tangent-to-identity formal series f, g ,*

$$T_N(f \circ g) \leq \max\{T_N(f), T_N(g)\}. \quad (6.2)$$

If $T_N(f) \neq T_N(g)$, equality holds, including when one type is infinite.

Proof. If both types are finite, take $A > \max\{T_N(f), T_N(g)\}$. After the common dilation $\mathcal{D}_A$, choose one C bounding the absolute nonlinear coefficients of both inputs by CN_n . Their composite is bounded coefficientwise in absolute value by $B_C \circ B_C$, hence by H_{2C} . Admissibility and Corollary 4.5 give type at most one after dilation. Since dilation commutes with composition, the undilated type is at most A . Let A decrease to the maximum. If an input type is infinite, the upper bound is automatic.

Suppose $T_N(f) > T_N(g)$. From $f = (f \circ g) \circ g^{(-1)}$, the already proved upper bound and exact inverse invariance give

$$T_N(f) \leq \max\{T_N(f \circ g), T_N(g)\}.$$

This forces $T_N(f \circ g) \geq T_N(f)$, also when $T_N(f) = \infty$. The other strict inequality is handled by $g = f^{(-1)} \circ (f \circ g)$. □

The estimate is not a product bound. Tangent dilations normalize both inputs at the same exponential scale, while every finite composition has only a subexponential overhead relative to the admissible envelope. The inverse identity then turns an upper bound into an exact law whenever the two scales differ.

6.2 Every allowed cancellation actually occurs

Theorem 6.3 (Complete type-only cancellation spectrum). *Fix two types $s, t \in [0, \infty]$. If $s \neq t$, every composite of tangent series of these types has type $\max(s, t)$. If $s = t = T$, the set of possible composite types is exactly $[0, T]$, with the natural extended-real interpretation for $T = \infty$.*

Proof. The unequal case and the upper restriction in the equal case follow from Theorem 6.2. Every finite positive type t is realized by the positive series

$$f_t(z) = z + \sum_{n \geq 1} t^n N_n z^{n+1}.$$

The identity realizes zero, and $z + \sum_{n \geq 1} n^n N_n z^{n+1}$ realizes infinity.

Let $0 \leq S < T \leq \infty$. Choose tangent f of type T and h of type S , and set $g = f^{(-1)} \circ h$. The unequal-type law and inverse invariance imply $T_N(g) = T$. Yet $f \circ g = h$ has type S . For

$S = T > 0$, choose $f = g$ with nonnegative nonlinear coefficients and type T . Since $f(z) \succeq z$, we have $f \circ f \succeq f$, so the composite has type at least T , and hence exactly T . This also works at infinity. At $T = 0$, the upper bound already forces zero. $\square$

Thus there is no missing universal scalar formula in the equal-type case: all values allowed by the upper bound occur. A more precise prediction must use information beyond the two types, such as signs, coefficient profiles, or a specified relation between the series.

6.3 Type-zero coordinates need not be analytic

Corollary 6.4 (Type-zero coordinate invariance). *If h is tangent to the identity and $T_N(h) = 0$, then for every tangent f ,*

$$T_N(h \circ f) = T_N(f \circ h) = T_N(f), \quad T_N(h^{\langle -1 \rangle} \circ f \circ h) = T_N(f). \quad (6.3)$$

Proof. Use the unequal-type law when $T_N(f) > 0$, and the maximum bound when it is zero. The inverse of h also has type zero, so the conjugation statement follows by applying the two identities successively. $\square$

Every analytic tangent germ has type zero because its coefficients have at most exponential growth while $N_n^{1/n} \rightarrow \infty$. But the corollary is strictly more general. For example,

$$h(z) = z + \sum_{n \geq 1} \sqrt{N_n} z^{n+1} \quad (6.4)$$

has radius of convergence zero, since $(\sqrt{N_n})^{1/(n+1)} \rightarrow \infty$, whereas

$$T_N(h) = \limsup_n N_n^{-1/(2n)} = 0.$$

It is therefore a genuinely divergent type-zero coordinate change under which every exact N -type is preserved.

6.4 Nonunit linear coefficients

For a formal series with zero constant coefficient and nonzero linear coefficient, use the same definition (2.2) on its nonlinear part. Multiplying the whole series by a fixed nonzero scalar does not change the type. Precomposing by $z \mapsto bz$ multiplies type by $|b|$.

Corollary 6.5 (Scalar coordinate scaling). *Let $F'(0) = a \neq 0$ and $G'(0) = b \neq 0$. Then*

$$T_N(F^{\langle -1 \rangle}) = |a|^{-1} T_N(F), \quad (6.5)$$

$$T_N(F \circ G) \leq \max\{|b| T_N(F), T_N(G)\}. \quad (6.6)$$

Equality holds in (6.6) whenever the two entries of the maximum differ. If h, k are invertible type-zero series with $h'(0) = b$, then

$$T_N(k \circ F \circ h) = |b| T_N(F). \quad (6.7)$$

Proof. Write $F = af$ and $G = bg$ with f, g tangent. Then $F^{\langle -1 \rangle} = (f^{\langle -1 \rangle})(z/a)$, proving (6.5). Also

$$F \circ G = ab(f_b \circ g), \quad f_b(z) = b^{-1}f(bz), \quad T_N(f_b) = |b| T_N(F).$$

Apply Theorem 6.2. Applying the result twice with a type-zero outer or inner series proves (6.7); when the surviving type is zero, the upper bound itself gives equality. $\square$

6.5 A filtered formal group

Let $\mathcal{G}_{\leq t}$ be the tangent formal series of type at most t . Exact inversion and the maximum bound show that each $\mathcal{G}_{\leq t}$ is a group under composition. The function

$$d(f, g) = T_N(f \circ g^{(-1)}) \quad (6.8)$$

is a right-invariant extended ultrapseudometric:

$$d(f, g) = d(g, f), \quad d(f, h) \leq \max\{d(f, g), d(g, h)\}, \quad d(f \circ k, g \circ k) = d(f, g).$$

Indeed the first identity follows from inversion invariance, the second from composition, and the third from cancellation of $k \circ k^{(-1)}$. It is a pseudometric rather than a metric because nontrivial type-zero series have zero distance from the identity. No left-invariance or normality of the type-zero subgroup is asserted.

7 Formal maps in several variables and Banach spaces

7.1 The coefficient norm must be specified

Let E be a real or complex Banach space. A formal map tangent to the identity is a sequence of bounded symmetric multilinear maps $A_m : E^m \rightarrow E$, written

$$F(x) = x + \sum_{m \geq 2} A_m(x, \dots, x). \quad (7.1)$$

No convergence of the displayed series is required. The coefficient norm is the multilinear operator norm

$$\|A_m\| = \sup_{\|x_1\|, \dots, \|x_m\| \leq 1} \|A_m(x_1, \dots, x_m)\|. \quad (7.2)$$

Define

$$T_N(F) = \limsup_{n \rightarrow \infty} \left(\frac{\|A_{n+1}\|}{N_n} \right)^{1/n}. \quad (7.3)$$

The norm in (7.2) is not silently replaced by the supremum of the diagonal polynomial on the unit ball. Polarization comparisons can have degree-dependent constants; exact exponential types must therefore keep the norm convention explicit.

For an m -linear map L , write

$$\text{Sym } L(x_1, \dots, x_m) = \frac{1}{m!} \sum_{\sigma \in \mathfrak{S}_m} L(x_{\sigma(1)}, \dots, x_{\sigma(m)}).$$

Then $\|\text{Sym } L\| \leq \|L\|$. Normalized symmetrization, rather than an unnormalized sum, is essential here.

7.2 Scalar majorants survive multilinear contraction

Suppose $G(x) = \sum_{j \geq 1} G_j(x, \dots, x)$ has zero constant term. The degree- m coefficient in $F \circ G$ is

$$(F \circ G)_m = \sum_{k=1}^m \sum_{\substack{j_1 + \dots + j_k = m \\ j_i \geq 1}} \text{Sym}(A_k(G_{j_1}, \dots, G_{j_k})), \quad (7.4)$$

where $A_1 = \text{id}$ for a tangent map. In each summand the m arguments are first divided into consecutive blocks of sizes j_i , then fed into the corresponding G_{j_i} , then into A_k ; the result is

symmetrized. On the diagonal, expansion of the polynomial composition gives exactly this formula.

The norm of a summand is at most

$$\|A_k\| \prod_{i=1}^k \|G_{j_i}\|. \quad (7.5)$$

There is no factor depending on $\dim E$ and no extra factor $m!$. Thus scalar series with nonnegative coefficient bounds majorize all composition coefficients in this convention.

Lemma 7.1 (Formal inverse and its scalar majorant). *A formal map (7.1) has a unique two-sided formal inverse G tangent to the identity. If $\|A_{n+1}\| \leq CN_n$, then*

$$\|G_{n+1}\| \leq h_{n+1}(C) \quad (n \geq 1). \quad (7.6)$$

Proof. Set $G_1 = \text{id}$. At degree $m \geq 2$, the equation $F \circ G = \text{id}$ requires

$$G_m = - \sum_{k=2}^m \sum_{j_1+\dots+j_k=m} \text{Sym}(A_k(G_{j_1}, \dots, G_{j_k})). \quad (7.7)$$

Every j_i in the sum is at most $m-1$, so the formula is a finite recursion defining a bounded symmetric coefficient. Formal composition is associative because every coefficient identity is an identity of finite polynomial substitutions. The same recursion gives a right inverse for G ; associativity then makes it equal to F , proving that the inverse is two-sided.

Taking norms in (7.7), applying (7.5), and inducting on m yields the scalar coefficient recursion for $H_C = z + CW_N(H_C)$. This proves (7.6). $\square$

7.3 Exact tangent calculus without a dimension loss

Theorem 7.2 (Banach-space zero-loss calculus). *Let N be admissible. For formal maps tangent to the identity on any real or complex Banach space, with norm (7.2),*

$$T_N(F^{(-1)}) = T_N(F), \quad (7.8)$$

$$T_N(F \circ G) \leq \max\{T_N(F), T_N(G)\}. \quad (7.9)$$

The composition inequality is an equality when the two types differ. Type-zero tangent precomposition, postcomposition, and conjugation preserve type.

Proof. Dilation $\mathcal{D}_A F(x) = AF(x/A)$ divides a homogeneous coefficient of degree $n+1$ by A^n , for positive real A . If the type is finite, normalize it with any larger A and use Lemma 7.1 and $T_N(H_C) = 1$ to obtain the inverse upper bound. Let A decrease to the type, and apply the bound to the inverse to get equality, exactly as in the scalar proof. Infinite types are handled by the same involution argument.

For composition, (7.4) transfers the scalar majorant $B_C \circ B_C \preceq H_{2C}$ to the multilinear norms. Dilation gives the maximum bound. Associativity, the inverse equality, and the identities used in Theorem 6.2 give equality for unequal types. The type-zero assertions follow in the same way. $\square$

The theorem concerns formal coefficient norms. It is not an analytic inverse-function theorem on an open ball, nor does it assert that a formal map is the Taylor expansion of a smooth map in a prescribed class. The scalar majorant argument removes a combinatorial exponential loss; it does not supply convergence where none was assumed.

7.4 A coefficient-sum convention in finite dimensions

There is a second useful norm convention for direct computation. For a homogeneous polynomial map $P : \mathbb{C}^d \rightarrow \mathbb{C}^d$ of degree m , put

$$\|P\|_{\text{coeff}} = \max_{1 \leq r \leq d} \sum_{|\alpha|=m} |[x^\alpha]P_r(x)|. \quad (7.10)$$

Multiplying scalar polynomials multiplies coefficient-sum majorants. For every monomial of the outer component, replacing all input variables by one common scalar majorant bounds its entire substitution. Summing the outer coefficients and taking the maximum over components therefore proves the same scalar composition bound as (7.5). The inverse recursion and Theorem 7.2 consequently hold for the type defined using (7.10) as well.

This is the convention used by the finite bivariate checks. We do not identify it with the multilinear operator norm or transfer exact types between norm conventions by an unstated degree-dependent equivalence. The two theorems follow from their respective scalar majorization properties.

7.5 General derivatives: sharp bounds, not a single scalar law

Theorem 7.3 (Sharp Jacobian bounds). *Let N be admissible and let a formal map F on a nonzero Banach space have bounded invertible linear coefficient A . Then*

$$\boxed{\frac{T_N(F)}{\|A\|} \leq T_N(F^{\langle -1 \rangle}) \leq \|A^{-1}\| T_N(F)}. \quad (7.11)$$

Both endpoints are sharp even in dimension two. In general the matrix A and the scalar value $T_N(F)$ do not determine $T_N(F^{\langle -1 \rangle})$.

Proof. Multiplication of all output coefficients by a fixed bounded invertible operator does not change type, since its norm and inverse norm are constant factors whose n th roots tend to one. Precomposition by a nonzero bounded linear operator B gives

$$\|P_{n+1}(B, \dots, B)\| \leq \|P_{n+1}\| \|B\|^{n+1},$$

and hence multiplies type by at most $\|B\|$.

Write $F = A \circ f$ with f tangent. Then $F^{\langle -1 \rangle} = f^{\langle -1 \rangle} \circ A^{-1}$, so

$$T_N(F^{\langle -1 \rangle}) \leq \|A^{-1}\| T_N(f^{\langle -1 \rangle}) = \|A^{-1}\| T_N(f) = \|A^{-1}\| T_N(F).$$

Apply this inequality to $F^{\langle -1 \rangle}$, whose linear coefficient is A^{-1} , to obtain the lower bound.

For sharpness, take $E = \mathbb{C}^2$ with the maximum norm and $A = \text{diag}(a, b)$, where $0 < a < b$. Let φ be a scalar tangent series of finite positive type T , and define

$$F_1(x, y) = (a\varphi(x), by), \quad F_2(x, y) = (ax, b\varphi(y)).$$

Both have derivative A and type T . Their inverses are

$$F_1^{\langle -1 \rangle}(x, y) = (\varphi^{\langle -1 \rangle}(x/a), y/b), \quad F_2^{\langle -1 \rangle}(x, y) = (x/a, \varphi^{\langle -1 \rangle}(y/b)),$$

and have types T/a and T/b , respectively. Since $\|A^{-1}\| = 1/a$ and $\|A\| = b$, the endpoints are attained with the same derivative and the same scalar input type. $\square$

The obstruction is directional: the growing nonlinear coefficients can be placed in different eigendirections. An exact general-Jacobian theory must therefore retain directional or multihomogeneous growth data, not merely improve a combinatorial estimate in a scalar norm.

8 Explicit weights separating the possible behaviors

The examples in this section require no undecided asymptotic input. Their envelopes are computed exactly. Together they show that the classification is neither a disguised log-convexity condition nor a consequence of root growth alone.

8.1 A staircase weight beyond every log-convex approximation

Theorem 8.1 (Non-convex admissibility). *The weight*

$$N_0 = 1, \quad N_n = 2^{n \lfloor \log_2 n \rfloor} \quad (n \geq 1) \quad (8.1)$$

is supermultiplicative and admissible. There is no positive eventually log-convex sequence V satisfying $\log(V_n/N_n) = o(n)$.

Proof. The roots $r_n = N_n^{1/n} = 2^{\lfloor \log_2 n \rfloor}$ are nondecreasing and tend to infinity. For any composition of n ,

$$\prod_i N_{j_i} = \prod_i r_{j_i}^{j_i} \leq r_n^{\sum_i j_i} = N_n.$$

Thus $S_n = N_n$, proving supermultiplicativity and admissibility.

Write $a_n = \log_2 N_n = n \lfloor \log_2 n \rfloor$. At $n = 2^m$ with $m \geq 1$, direct calculation gives

$$2a_n - a_{n-1} - a_{n+1} = n - 1. \quad (8.2)$$

Indeed $a_{n-1} = (n-1)(m-1)$, $a_n = nm$, and $a_{n+1} = (n+1)m$. If $\log_2 V_n = a_n + e_n$ with $e_n = o(n)$, then $2e_n - e_{n-1} - e_{n+1} = o(n)$ along these indices. The convexity defect for $\log_2 V$ is therefore $n - 1 + o(n) > 0$, contradicting eventual log-convexity, which requires that defect to be at most zero. $\square$

This disproves the necessity of the log-convex-envelope route suggested as one possible approach in the repository's arbitrary-weight question [1]. Subexponential equivalence to a supermultiplicative sequence is necessary and sufficient; subexponential equivalence to a log-convex sequence is strictly more restrictive.

8.2 Admissibility without asymptotically nondecreasing roots

Proposition 8.2 (Persistent downward root jumps). *The weight*

$$N_0 = 1, \quad N_n = 2^{n^2 + (-1)^n n} \quad (n \geq 1) \quad (8.3)$$

is supermultiplicative and admissible. At every even n ,

$$\frac{N_{n+1}^{1/(n+1)}}{N_n^{1/n}} = \frac{1}{2}. \quad (8.4)$$

It is not subexponentially equivalent to a sequence whose roots are eventually nondecreasing.

Proof. Let $a_n = n^2 + (-1)^n n$. For positive u, v , the difference $a_{u+v} - a_u - a_v$ equals $2uv$ when both are even, equals $2u(v-1)$ when u is even and v odd, and equals $2uv + 2(u+v)$ when both are odd. The remaining mixed case is symmetric. Each difference is nonnegative, so N is supermultiplicative. Its roots are $2^{n+(-1)^n} \rightarrow \infty$, proving admissibility and (8.4).

If V were subexponentially equivalent to N , then $V_n^{1/n}/N_n^{1/n} \rightarrow 1$. The ratio of successive roots of V along even n would tend to $1/2$, contradicting eventual nondecrease. $\square$

Hence even a root-monotonicity substitute, up to vanishing multiplicative error in the roots, is not necessary. The partition-product condition is genuinely more flexible.

8.3 Every finite distortion is realized

Theorem 8.3 (A one-parameter finite-loss family). *For $c > 1$, define*

$$N_0 = 1, \quad N_n = n! c^{-n \mathbf{1}_{\{n \text{ odd}\}}}. \quad (8.5)$$

Then $N_n^{1/n} \rightarrow \infty$, and its envelope is

$$S_n = \begin{cases} n!, & n \text{ even}, \\ \max\{n!c^{-n}, (n-1)!/c\}, & n \text{ odd}. \end{cases} \quad (8.6)$$

In particular, $\Delta(N) = \mathcal{I}(N) = c$. The series $z - W_N$ has type one and inverse type exactly c .

Proof. The roots tend to infinity because $n!$ has superexponential roots and the parity factor reduces them by at most c . One elementary lower bound is $n! \geq \lfloor n/2 \rfloor^{n/2}$.

For every composition of n , the product of the part factorials is at most $n!$, and all powers of c^{-1} reduce it. If n is even, the one-part composition attains $n!$. If n is odd, a nontrivial composition contains an odd part and hence a penalty of at least c^{-1} . Also every nontrivial composition has factorial product at most $(n-1)!$. To see this, group parts to two positive integers $a, n-a$; then $a!(n-a)! = n!/ \binom{n}{a} \leq (n-1)!$. The product before grouping is no larger. Therefore every nontrivial product is at most $(n-1)!/c$, attained by the partition $(n-1, 1)$ for odd $n \geq 3$. The one-part product is $n!c^{-n}$. At $n=1$ the displayed maximum has the same value $1/c$, so the formula remains valid.

For even n , $S_n/N_n = 1$. For odd n ,

$$\frac{S_n}{N_n} = \max \left(1, \frac{c^{n-1}}{n} \right),$$

whose n th root tends to c along the odd indices. Apply Theorem 2.3 and Corollary 4.5. $\square$

A finite distortion larger than one does not necessarily destroy the zero-type class: (2.11) still preserves type zero. It does destroy exact positive type. The next example demonstrates a stronger failure unavailable when the distortion is finite.

8.4 Deep dips and an exact supermultiplicative repair

Theorem 8.4 (An infinite-distortion weight). *Let*

$$N_0 = 1, \quad N_n = \begin{cases} 2^{n^2}, & n \text{ even}, \\ 2^{(n^2-1)/4}, & n \text{ odd}. \end{cases} \quad (8.7)$$

Then $N_n^{1/n} \rightarrow \infty$, but its envelope is

$$S_n = \begin{cases} 2^{n^2}, & n \text{ even}, \\ 2^{(n-1)^2}, & n \text{ odd}, \end{cases} \quad (8.8)$$

and $\Delta(N) = \infty$.

Proof. The root exponents are n on even indices and $(n^2-1)/(4n)$ on odd indices, so both tend to infinity. An odd part $j \geq 3$ in a partition may be replaced by the even part $j-1$ and the part one without decreasing the product, because $N_1 = 1$ and

$$(j-1)^2 - \frac{j^2-1}{4} = \frac{(3j-5)(j-1)}{4} \geq 0.$$

Parts equal to one are left alone. After all such replacements, combine the even parts into one even part: their squared exponents have sum at most the square of their sum. If the original total n is even, the resulting product is at most 2^{n^2} , attained by the one-part partition. If n is odd, at least one part one remains, so the product is at most $2^{(n-1)^2}$, attained by $(n-1, 1)$, or by (1) when $n = 1$.

At odd n ,

$$\frac{1}{n} \log_2 \frac{S_n}{N_n} = \frac{3}{4}n - 2 + \frac{5}{4n} \longrightarrow \infty.$$

Thus the envelope repair has an unbounded exponential defect. $\square$

Theorem 8.5 (Type zero can invert to infinite type). *For the weight (8.7), define*

$$f(z) = z - z^2 - \sum_{\substack{n \geq 2 \\ n \text{ even}}} 2^{n^2 - n\lceil\sqrt{n}\rceil} z^{n+1}. \quad (8.9)$$

Then $T_N(f) = 0$ and $T_N(f^{(-1)}) = \infty$.

Proof. At even n , the normalized n th root of the coefficient of f is $2^{-\lceil\sqrt{n}\rceil} \rightarrow 0$; at odd $n \geq 3$ it is zero. The single quadratic coefficient does not affect type. Hence $T_N(f) = 0$.

Write the negative nonlinear coefficients of f as $-b_{n+1}$ with $b_{n+1} \geq 0$. Its inverse has nonnegative coefficients by Lemma 3.2. For odd $n \geq 3$, the two ordered compositions $(n-1, 1)$ and $(1, n-1)$ in the $k = 2$ term yield

$$[z^{n+1}]f^{(-1)} \geq (n+2) 2^{(n-1)^2 - (n-1)\lceil\sqrt{n-1}\rceil}. \quad (8.10)$$

Indeed the Lagrange factor is $\binom{n+2}{2}/(n+1) = (n+2)/2$, and there are two distinct orderings; also $b_2 = 1$. After division by N_n , the base-two logarithm of the normalized n th root is bounded below by

$$\frac{\log_2(n+2) + (n-1)^2 - (n-1)\lceil\sqrt{n-1}\rceil - (n^2 - 1)/4}{n} = \frac{3}{4}n - \sqrt{n} + O(1),$$

which tends to infinity along odd n . This proves the assertion. $\square$

The mechanism is explicit: an even degree with a large weight mixes with a single quadratic term and lands in a much smaller odd-degree weight. The failure is not an oscillatory sign phenomenon; all inverse contributions used in the lower bound are positive.

8.5 Harmless oscillations can be very irregular locally

As a simple contrast, take

$$N_n = (n+1)!, \quad V_n = (n+1)! 2^{(-1)^n \lfloor \sqrt{n} \rfloor} \quad (n \geq 1), \quad (8.11)$$

with zeroth terms one. Proposition 5.5 makes N admissible. Since $\log(V_n/N_n) = O(\sqrt{n}) = o(n)$, Corollary 5.2 makes V admissible with exactly the same types. No direct check of adjacent quotients or log-convexity of V is required. Together with the preceding examples, this emphasizes that the relevant scale is the envelope defect divided by n , not the visible size of local oscillations alone.

9 Consequences for coefficient transforms and transseries

9.1 Exact radii of weight-normalized transforms

For a scalar formal series with nonlinear coefficients f_{n+1} , define its explicitly normalized transform

$$\mathcal{B}_N f(\zeta) = \sum_{n \geq 1} \frac{f_{n+1}}{N_n} \zeta^n. \quad (9.1)$$

This definition is merely a coefficient normalization. It is not, for a general N , a claim that an associated Laplace kernel or a summation procedure exists.

The radius of convergence $R_N(f)$ is

$$R_N(f) = \frac{1}{T_N(f)}, \quad (9.2)$$

with $1/0 = \infty$ and $1/\infty = 0$. One can verify the formula directly: inside the reciprocal limsup radius the terms are eventually bounded by a convergent geometric series; outside it, a subsequence of terms fails to tend to zero.

Corollary 9.1 (Radius preservation and optimal radius distortion). *For admissible N , every tangent formal series satisfies*

$$R_N(f^{\langle -1 \rangle}) = R_N(f). \quad (9.3)$$

If $N_n^{1/n} \rightarrow \infty$ and $1 \leq \Delta(N) < \infty$, then for $0 < R_N(f) < \infty$,

$$\frac{R_N(f)}{\Delta(N)} \leq R_N(f^{\langle -1 \rangle}) \leq \Delta(N) R_N(f). \quad (9.4)$$

The constant is sharp. Radii zero and infinity are individually preserved when $\Delta(N) < \infty$.

Proof. Take reciprocals in Theorems 2.2 and 2.3, with the endpoint cases handled by the zero- and infinite-type statements. The negative extremal input has radius one and inverse radius $1/\Delta(N)$. $\square$

For log-convex factorial or refined factorial coefficient scales this recovers the radius interpretation that motivates the predecessor [1]. The extension is that the same interpretation now has an exact necessary-and-sufficient criterion for arbitrary positive scales, including the non-convex examples above.

9.2 Where this fits in a transseries calculation

An integer-indexed sector of a transseries calculation can produce a formal series in a small parameter q , with a chosen coefficient scale. When its normalization is tangent to the identity, the results here apply to the reversion in q without requiring the series to converge. They also apply to finite systems of such formal series through Section 7. The envelope is an explicit preprocessing step: compute or estimate S_n , and distinguish whether its defect is sublinear, linear with finite limsup, or unbounded on the linear scale.

For example, a coefficient calculation known to have finite exact type in the staircase scale (8.1) can be inverted with precisely the same type even though no equivalent log-convex scale exists. In the family (8.5), a generic membership statement obscures a real factor c in the worst possible transformed radius. The deep-dip example warns that a superexponential normalization by itself can hide an arbitrarily severe nonlinear growth instability.

These statements should not be extrapolated to an arbitrary Hahn support by replacing the integer n with an exponent. The proof uses finite compositions of an integer and a concrete count of the possible lengths. Accumulating action sets require a separate support-summability and counting theory. Similarly, no Stokes constant, angular continuation, Borel summability direction, or optimal truncation error follows merely from the radius in (9.2).

10 Finite algorithms and executed verification

10.1 What a finite prefix can certify

For rational input weights $N_1, \dots, N_d$, the recurrence (3.1) computes $S_1, \dots, S_d$ and records a maximizing composition for each degree. It uses $O(d^2)$ rational multiplications and comparisons. This is an arithmetic-operation count, not a polynomial bit-complexity claim: the numerators and denominators may be very large.

A second computation uses the ordered Lagrange formula, precomputing truncated powers of $\sum_{j \geq 1} N_j z^j$. An independent implementation enumerates unordered partitions and evaluates (3.5). Agreement of these paths checks the coefficient index, multiplicity factor, and sign convention. A third implementation computes a fixed number of formal Picard iterations and checks the inverse equations. It does not use the partition formula.

For a chosen rational η and C , an exact integer B is found by binary-searching the inequalities $B^j \geq N_j$ over the required finite prefix. The quantities P, Q, R of Theorem 4.2 are then rational, so the certificate can be tested with no transcendental arithmetic. Finite tests cannot verify that $N_n^{1/n} \rightarrow \infty$ or that a limsup equals one; those assertions are proved from the explicit hypotheses in the article.

10.2 Recorded checks

The accompanying standard-library Python program was executed with Python 3.13.5, seed 20260929, and maximum excess degree 36 (ordinary coefficient degree 37). It reported **5,443 counted exact finite checks**, all passing. The counter names and precise scope are preserved in `data/verification.json`; a summary follows.

Check family	Count
Envelope witnesses and positive inverse lower bounds	216
Independent partition-formula comparisons	108
Exact sparse–dense upper certificates	648
Envelope supermultiplicativity inequalities	3,780
Six weight families: two-sided inverse and Picard audits	6
Explicit envelopes: two supermultiplicative families	72
Deep-dip and finite-loss envelope formulas	72
Additional arbitrary-weight finite audits	144
Signed coefficient majorants and dilation checks	288
Signed two-sided inverse cases	24
Type-zero-to-infinite-type finite lower bounds	17
Bivariate homogeneous coefficient-norm bounds	48
Bivariate two-sided inverse cases	8
Staircase convexity-defect identities	12
Total	5,443

The six structured families are the shifted factorial weight, the staircase weight, the alternating supermultiplicative weight, finite loss with $c = 2$, deep dips, and the subexponential factorial

perturbation. Their positive inverse coefficients and envelopes were computed through excess degree 36. Independent partition enumeration was checked through excess degree 18. The two scalar inverse compositions were checked through ordinary degree 16 for these families, with independent Picard checks through degree 12. Twenty-four seeded signed-series cases were checked through ordinary degree 13.

Eight seeded bivariate maps were inverted through total degree 7. Both compositions were checked against the identity, and the homogeneous coefficient-sum norms (7.10) were compared with the scalar inverse majorant. These are finite rational polynomial tests; the infinite-dimensional statement follows from the multilinear proof, not from sampling finite-dimensional cases.

Every asserted identity and inequality in the verification program uses integers or rational numbers. The logarithms in the diagnostic tables below use ordinary floating-point arithmetic, are labeled accordingly, and are not interval certificates. A counted check may aggregate several coefficient comparisons; the total is an audit counter, not a count of independent mathematical theorems.

10.3 A diagnostic separating envelope defect from tree overhead

The finite observations are easiest to interpret after separating

$$d_n = \frac{1}{n} \log_2(S_n/N_n), \quad r_n = \frac{1}{n} \log_2(h_{n+1}(1)/S_n). \quad (10.1)$$

The theorem predicts $r_n \rightarrow 0$ for each of the displayed weights. The envelope defect d_n behaves very differently: it is zero for the staircase, tends to one along odd indices in the $c = 2$ finite-loss family, and tends to infinity along odd indices for deep dips.

Weight	Excess degree n	d_n	r_n
Staircase	9	0.000000	0.154819
Staircase	17	0.000000	0.066476
Staircase	25	0.000000	0.138526
Staircase	33	0.000000	0.032306
Deep dips	9	4.888889	0.384381
Deep dips	17	10.823529	0.249878
Deep dips	25	16.800000	0.190196
Deep dips	33	22.787879	0.155433
Finite loss, $c = 2$	9	0.536675	0.507991
Finite loss, $c = 2$	17	0.700737	0.273671
Finite loss, $c = 2$	25	0.774246	0.200121
Finite loss, $c = 2$	33	0.816837	0.161217

These are rounded diagnostics from exact rational coefficients, not certified logarithmic intervals. In particular, the staircase values of r_n are visibly nonmonotone. The proofs require no monotone convergence and do not infer the limits from the displayed finite degrees.

10.4 Reproduction and output discipline

The package contains `article.tex`, its compiled PDF, the verification code, recorded JSON and CSV data, provenance and proof-audit notes, and build commands. The article embeds its bibliography and table; it compiles without reading the verification outputs or accessing the network. From the package directory, run

```
python3 code/verify.py --order 36
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verifier’s default output is `build/verification/`, leaving `data/` unchanged. An explicit `--output` argument can select another destination. It requires no third-party Python package, contacts no external service, and does not modify the upstream repository. PDF build and rendering checks are recorded separately from the mathematical verification report.

11 Proof dependencies and a formalization route

11.1 The short dependency chain

The universal classification rests on a particularly small chain of arguments. First, finite Lagrange inversion converts inverse coefficients to positive partition products. Second, replacing many small parts by one large part proves that maximizing partitions have sublinear length. Third, a sparse–dense decomposition bounds every product relative to the supermultiplicative envelope and separates a small entropy cost from an arbitrarily strong dense penalty. These three steps prove (2.14). Dilation, absolute values, and the inverse involution then prove the exact distortion and zero-loss criterion.

Composition introduces only the majorant $B_C \circ B_C \preceq H_{2C}$. Unequal-type equality and all cancellation examples use formal group identities. The Banach extension requires the norm inequality for normalized symmetrized contractions, after which the same scalar majorants apply. No compactness theorem for smooth functions, analytic continuation theorem, or resurgent input is hidden in the chain.

11.2 Finite algebra before asymptotics

A practical formalization can begin with finite words of positive natural numbers and their sum. The envelope recurrence, concatenation inequality, and least-majorant property are finite statements about maxima. They can be separated from formal power series entirely. The next layer is the finite coefficient identity (3.4), together with its absolute-value majorant and the equivalence to multiplicity-vector sums.

The finite sparse–dense certificate is a useful standalone target. It requires counting positive compositions and tracking the total mass of parts at most L , but contains no limit. Formalizing it first isolates the only combinatorial estimates needed for the asymptotic theorem. The lower bound from a maximizing word should also be stated separately, so that the $C < 1$ case is not accidentally lost.

11.3 Limits and extended-real endpoints

The analytic layer can be divided into the sparse-maximizer lemma, the binomial entropy estimate, the dense-suppression limit, and the central root limit. One must preserve the order of quantifiers: first fix η , then fix B from finitely many weights, then let $n \rightarrow \infty$, and only afterwards let $\eta \downarrow 0$. The assertion is not uniform in all $C > 0$ simultaneously; each C is fixed before the limit.

The type layer should make zero and infinity explicit. A convenient approach is to prove finite coefficient bounds for every $A > T_N(f)$, derive the upper inequality when the input type is finite, and handle infinite types by applying the same inequality to the inverse. The polynomial obstruction deserves its own lemma because it supplies the necessity of a full root limit, not merely an unbounded subsequence.

11.4 Vector-valued coefficients and what has not been checked

For the Banach extension, a representation by symmetric continuous multilinear coefficients avoids an implicit polarization loss. The formalization needs normalized symmetrization, contraction, finite formal composition, and the recursively defined inverse. Associativity can be proved coefficientwise, independently of any convergent power-series infrastructure.

No new Lean code is supplied or claimed to compile in this package. The proposed stages are a dependency plan, not a statement that matching library declarations already exist. The executed rational verification program can provide regression examples after the formal statements are implemented, but it does not certify the universal theorems.

12 Further research questions and concrete next targets

The arbitrary-weight scalar zero-loss classification and the scalar composition spectrum are settled by the proofs in this article. The following questions concern additional information not encoded by those answers. They are proposed research directions; not every item is claimed to be a previously published open problem.

12.1 Optimal subexponential overhead

Theorem 2.4 proves $\log h_{n+1}(C) - \log S_n = o(n)$, while Theorem 4.2 supplies an explicit but generally nonoptimal finite upper bound. Determine the smallest attainable subexponential overhead from a specified weight profile. A concrete first target is to express it in terms of the number and entropy of partitions whose products lie within a chosen factor of S_n . This would refine the repository’s question on optimal reversion cost without presuming log-convex local quotient asymptotics.

12.2 Anisotropic exact types under a general Jacobian

Theorem 7.3 proves that one scalar type and an invertible matrix are insufficient for a unique inverse type. Define a directional or multihomogeneous type profile that transforms exactly under linear precomposition and is preserved, in an appropriate sense, by tangent inversion. A useful finite-dimensional target is a profile indexed by polyradii, together with an exact covariance law under diagonal Jacobians. Non-diagonal changes would then test how much angular data must be retained.

12.3 Exact composition distortion outside the zero-loss class

For a nonadmissible superexponential weight with finite $\Delta(N)$, the same positive majorant gives an upper bound $T_N(f \circ g) \leq \Delta(N) \max(T_N(f), T_N(g))$. Is this constant optimal? A single composition has a restricted two-level substitution structure, whereas an inverse permits arbitrarily deep structures. Identify the correct restricted envelope and determine whether its defect can be strictly smaller than the inversion defect. This asks for a new classification, not a repetition of the zero-loss case.

12.4 Effective criteria for presented weights

The finite recurrence computes every prefix of S , but the condition $\log(S_n/N_n) = o(n)$ is asymptotic. For which explicitly presented families can it be decided or certified effectively? Candidate classes include piecewise-polynomial logarithmic weights on residue classes, regularly

varying weights with controlled oscillation, and weights given by finite max-plus recurrences. An algorithm should output a modulus of the envelope defect, not extrapolate it from a long finite prefix.

12.5 Quantitative cancellation profiles

Theorem 6.3 determines everything that two scalar types alone can say. What additional finite or asymptotic data predicts the output type in a structured family of equal-type inputs? A concrete target is a noncancellation theorem for prescribed sign cones or for coefficients concentrated on separated degree sets. Another is a stability theorem describing how a small coefficient perturbation moves a pair away from the exact inverse relation.

12.6 Differentiation, flows, and iterative logarithms

Admissibility was characterized for inversion and composition only. Determine the additional exact-type conditions needed for differentiation, formal flows, and the iterative logarithm of a tangent map. Adjacent weight ratios, absent from the main criterion, may reappear when an operator changes degree. A basic test is the staircase weight: its exact composition stability coexists with large local changes in its growth profile. No implication about these other operations is assumed here.

12.7 Hahn supports and accumulating actions

Replace the integer excess degree by an ordered action semigroup. One must specify which decompositions are finite, how their number is controlled, and what notion of a coefficient-growth scale is invariant under support refinements. A first target is a finitely generated positive semigroup with a fixed positive additive degree. Only after that case is understood should one address accumulating actions, where a finite integer-composition count is no longer available.

12.8 Realization and summation compatible with the envelope

For which admissible but non-log-convex weights do formal series admit useful analytic realizations with bounds preserving the exact type? Can a moment transform or acceleration procedure be built directly from the envelope S , and when does its sum commute with inversion? These questions require analytic continuation and growth hypotheses beyond coefficient bounds. The equality of normalized-transform radii in Corollary 9.1 supplies a necessary consistency check, not a summation theorem.

12.9 Finite certificates with bit-complexity bounds

The present dynamic program has a quadratic count of rational operations, while the integers in the examples can be huge. Develop compressed representations for logarithmic or factorial weights and prove bit-complexity bounds for envelope witnesses and sparse–dense certificates. A useful target is an independently checkable certificate whose size is small relative to the requested degree and the description length of the weight, rather than to its expanded numerator.

12.10 A proof-assistant formalization of the classification

The finite maximum-product closure and the exact certificate offer a manageable entry point. The main goal is a formally checked theorem quantifying over all positive weights, including the necessity direction and types zero and infinity. The Banach-space extension should then

reuse the scalar result through a single majorant interface. Formalizing just selected numerical examples would not establish the classification; formalizing the central root limit would be the decisive milestone.

Conclusion

The obstacle to exact formal inversion is not the absence of log-convexity. It is the exponential discrepancy between a weight and the products that additive partitions can form from it. Superexponential root growth suppresses the combinatorial cost of dense products, leaving the supermultiplicative envelope as the exact exponential-scale answer.

This observation yields a complete arbitrary-weight criterion, an optimal distortion constant, a canonical repair, and a sharp nonlinear type calculus. The examples show that every part of the criterion matters: strongly non-convex weights can work perfectly, finite parity penalties can create a prescribed loss, and deeper parity dips can turn a type-zero input into an inverse of infinite type. The same positive majorants survive in formal Banach-space maps once the coefficient norm is fixed precisely. What remains is not the basic scalar classification, but finer overheads, directional information, different support geometries, and analytic realization.

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